Bureau
International des
Poids et
Mesures

Le système international d'unités
The International System of Units

9 édition 2005

Annexe 2
Appendix 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.18
Part 2.18

Calcium ($\lambda \approx 657\,\mathrm{nm}$) (2005)

Calcium ($\lambda \approx 657\,\mathrm{nm}$) (2005)

Bureau International
des Poids et Mesures

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second Part 2.18: Calcium ($\lambda \approx 657\ \mathrm{nm}$) (2005)

9th edition 2005

Appendix 2

Absorbing atom ^{40}Ca, $^1S_0 - {}^3P_1$**;** $\Delta m_J = 0$ transition

1. CIPM recommended values

The values $f = 455\,986\,240\,494\,140$ Hz
$\lambda = 657\,459\,439.291\,683$ fm

with a relative standard uncertainty of 1.8×10^{-14} apply to the radiation of a laser stabilized to Ca atoms. The values correspond to the mean frequency of the two recoil-split components for atoms which are effectively stationary, i.e. the values are corrected for the second-order Doppler shift.

2. Source data

Adopted value $f = 455\,986\,240\,494\,140(8)$ Hz

$u_c/y = 1.8 \times 10^{-14}$

for which:

$\lambda = 657\,459\,439.291\,683(12)$ fm

$u_c/y = 1.8 \times 10^{-14}$

calculated from

f/Hz	u_c/y	source data
455 986 240 494 144	1.2×10^{-14}	[1]
455 986 240 494 135.8	7.5×10^{-15}	[2], [3]
Unweighted mean:	$f = 455\,986\,240\,494\,140$ Hz	

The CCL decided to adopt the unweighted mean of the two values, with an uncertainty of 8 Hz, equal to the difference between the values.

References

[1] Degenhardt C., Stoehr H., Lisdat Ch., Wilpers G., Schnatz H., Lipphardt B., Nazarova T., Pottie P.-E., Sterr U., Helmcke J., Riehle F., Calcium Optical Frequency Standard with Ultracold Atoms: Approaching 10^{-15} Relative Uncertainty, *Phys. Rev. A*, **72**, 062111/1-17, 2005.

[2] Wilpers G., Oates C. W., Hollberg L., Improved Uncertainty Budget for Optical Frequency Measurements with Microkelvin Neutral Atoms: Results for a High-Stability ^{40}Ca Optical Frequency Standard, *Appl. Phys. B* **85**, pp. 31-44, 2006.

[3] Wilpers G., Oates C. W., Diddams S. A., Bartels A., Fortier T M., Oskay W. H., Bergquist J. C., Jefferts S. R., Heavner T. P., Parker T. E., Hollberg L., Absolute Frequency Measurement of the Neutral ^{40}Ca Optical Frequency Standard at 657 nm based on Microkelvin Atoms, *Metrologia*, **44**, pp. 146-151, 2007.